



STUDENT DETAILS

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Admission Number: BS CCS/2024/42922

Course: Bachelor of Science in Computer Science

Unit Code: BIT 4138

Unit Name: Advanced Cryptography

Lecturer Name: Mr Michael Nyoro

Academic Year: May 2026

PROJECT DETAILS

Project Title: Digital Signature Implementation and Certificate Management

Selected Technologies:

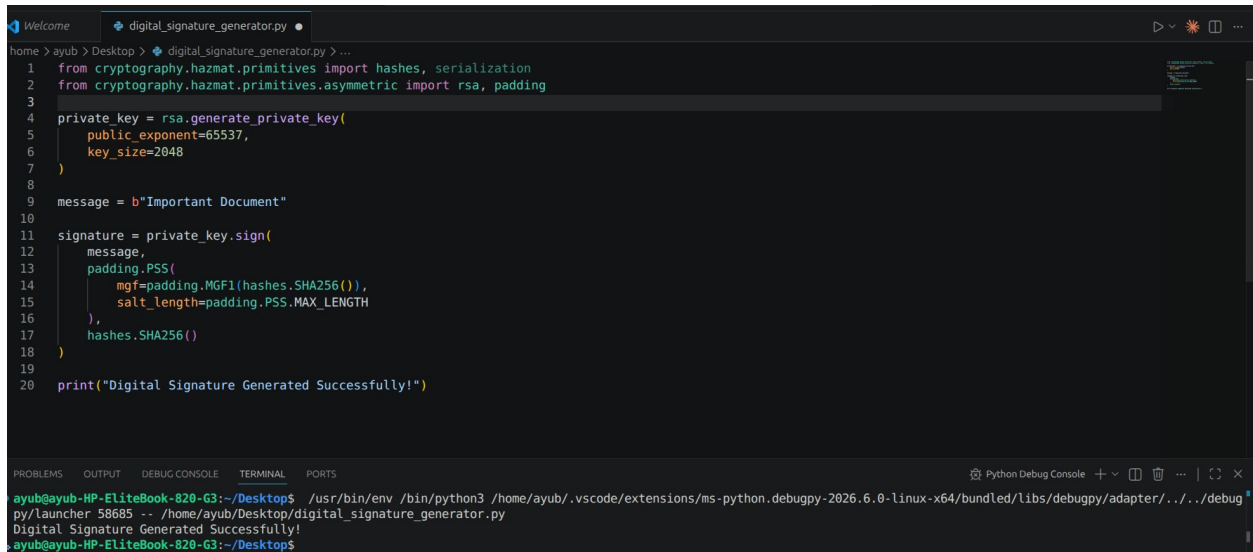
- Python
- Openssl

Github Link: https://github.com/jobcommits/BIT4138_Advanced_Cryptography

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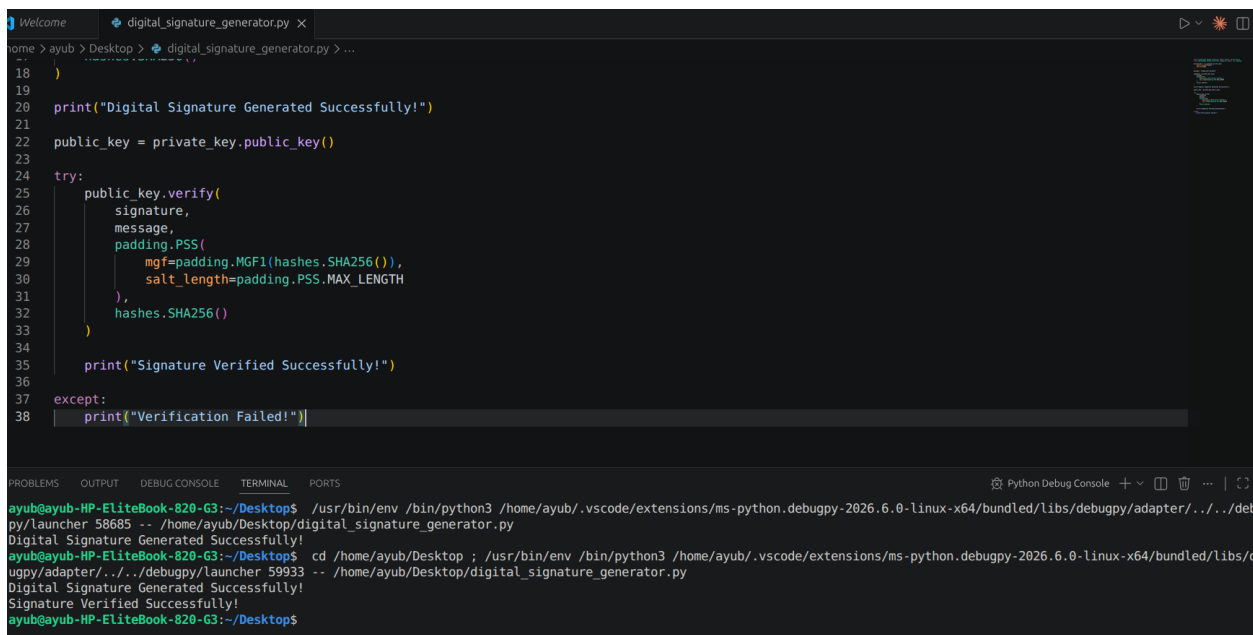
Week 7: Digital Signatures and Certificates



```
1 from cryptography.hazmat.primitives import hashes, serialization
2 from cryptography.hazmat.primitives.asymmetric import rsa, padding
3
4 private_key = rsa.generate_private_key(
5     public_exponent=65537,
6     key_size=2048
7 )
8
9 message = b"Important Document"
10
11 signature = private_key.sign(
12     message,
13     padding.PSS(
14         mgf=padding.MGF1(hashes.SHA256()),
15         salt_length=padding.PSS.MAX_LENGTH
16     ),
17     hashes.SHA256()
18 )
19
20 print("Digital Signature Generated Successfully!")
```

ayub@ayub-HP-EliteBook-820-G3:~/Desktop\$ /usr/bin/env /bin/python3 /home/ayub/.vscode/extensions/ms-python.debugpy-2026.6.0-linux-x64/bundled/libs/debugpy/adapter/../../debugpy/launcher 58685 -- /home/ayub/Desktop/digital_signature_generator.py
Digital Signature Generated Successfully!
ayub@ayub-HP-EliteBook-820-G3:~/Desktop\$

Figure 1: Digital Signature Generation



```
18
19
20 print("Digital Signature Generated Successfully!")
21
22 public_key = private_key.public_key()
23
24 try:
25     public_key.verify(
26         signature,
27         message,
28         padding.PSS(
29             mgf=padding.MGF1(hashes.SHA256()),
30             salt_length=padding.PSS.MAX_LENGTH
31         ),
32         hashes.SHA256()
33     )
34     print("Signature Verified Successfully!")
35
36 except:
37     print("Verification Failed!")
```

ayub@ayub-HP-EliteBook-820-G3:~/Desktop\$ /usr/bin/env /bin/python3 /home/ayub/.vscode/extensions/ms-python.debugpy-2026.6.0-linux-x64/bundled/libs/debugpy/adapter/../../debugpy/launcher 58685 -- /home/ayub/Desktop/digital_signature_generator.py
Digital Signature Generated Successfully!
ayub@ayub-HP-EliteBook-820-G3:~/Desktop\$ cd /home/ayub/Desktop ; /usr/bin/env /bin/python3 /home/ayub/.vscode/extensions/ms-python.debugpy-2026.6.0-linux-x64/bundled/libs/debugpy/adapter/../../debugpy/launcher 59933 -- /home/ayub/Desktop/digital_signature_generator.py
Digital Signature Generated Successfully!
Signature Verified Successfully!
ayub@ayub-HP-EliteBook-820-G3:~/Desktop\$

Figure 2: Signature Verification Process

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Debug Console + - [ ] ... | X
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$ openssl req -x509 -newkey rsa:2048 -keyout private.key -out certificate.crt -days 365 -nodes
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:KE
State or Province Name (full name) [Some-State]:Nairobi
Locality Name (eg, city) []:Nairobi
Organization Name (eg, company) [Internet Widgits Pty Ltd]:MKU
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:JOB
Email Address []:jobb.roman@gmail.com
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$
```

Figure 3: Certificate Generation Process

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Debug Console + - [ ] ... | X
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$ openssl x509 -in certificate.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      26:7c:a6:07:05:73:87:42:36:88:9b:38:ea:9b:d0:aa:f2:21:27:60
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C = KE, ST = Nairobi, L = Nairobi, O = MKU, OU = IT, CN = JOB, emailAddress = jobb.roman@gmail.com
    Validity
      Not Before: Jun 22 08:44:05 2026 GMT
      Not After : Jun 22 08:44:05 2027 GMT
    Subject: C = KE, ST = Nairobi, L = Nairobi, O = MKU, OU = IT, CN = JOB, emailAddress = jobb.roman@gmail.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:bb:b8:e2:49:95:a8:d7:34:c0:05:76:c6:e9:c5:
        b1:1d:cb:b9:3c:60:cb:93:09:5e:45:77:fd:1d:a8:
        37:12:2d:85:99:bc:35:dd:4a:47:46:0c:29:ca:31:
        7f:3a:ff:74:4b:43:7d:6e:a3:90:b5:be:7b:ab:e9:
        dc:3e:87:96:da:9c:34:e8:5a:d4:cb:fa:5a:88:43:
        c3:67:45:ad:93:7b:5e:26:96:dc:df:0f:29:1a:de:
        c2:94:79:03:d7:11:3c:83:0e:9a:e2:a3:30:c7:c1:
        4e:91:08:d5:c1:66:99:06:35:87:88:67:6c:6b:4f:
        35:d7:51:58:18:e9:c0:0b:28:bf:70:00:42:fd:08:
        9f:1c:df:14:cd:18:97:43:5c:35:a6:cd:15:5e:a7:
        5f:84:1f:4b:48:7f:38:b6:cf:af:d2:76:82:42:c5:
        4d:9c:9d:55:35:78:ac:be:41:f1:28:ff:06:0a:91:
        60:ed:90:69:73:d7:3c:32:cc:4d:f3:ce:bf:9a:5c:
        03:84:5e:78:20:0f:ce:f4:d6:66:e9:28:03:43:5c:
        f5:ca:a8:61:c6:a0:0e:64:bc:0c:13:12:b8:38:f3:
        20:7d:10:c5:bc:27:7b:6d:34:d9:49:a3:bb:42:d6:
        6a:52:c5:76:64:52:84:e9:87:8a:c5:b8:d9:95:40:
        6c:25
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Subject Key Identifier:
        3C:A8:77:5C:8D:B6:5D:E5:56:4C:36:A3:4C:67:78:C7:19:23:59:FE
      X509v3 Authority Key Identifier:
        3C:A8:77:5C:8D:B6:5D:E5:56:4C:36:A3:4C:67:78:C7:19:23:59:FE
      X509v3 Basic Constraints: critical
      CA:TRUE
```

Figure 4: Certificate contents 1

```
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$
```

```
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$
```

```
29         msg=padding.MGF1(hashes.SHA256()),
30         salt_length=padding.PSS.MAX_LENGTH
31     ),
32     hashes.SHA256()
33 )
34
35 print("Signature Verified Successfully!")
36
37 except:
38     print("Verification Failed!")
39
40 document = b"Important Document"
41
42 try:
43     public_key.verify(
44         signature,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Debug Console

```
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$ openssl x509 -in certificate.crt -text -noout
2e:0f:18:f1:f2:10:95:66:df:3d:32:7e:30:d7:8a:df:d3:11:
76:62:b1:58
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$ cd /home/ayub/Desktop ; /usr/bin/env /bin/python3 /home/ayub/.vscode/extensions/ms-python.debugpy-2026.6.0-linux-x64/bundled/Libs/debugpy/adapter/../../debugpy/launcher 57713 -- /home/ayub/Desktop/digital_signature_generator.py
Digital Signature Generated Successfully!
Signature Verified Successfully!
Document is Authentic.
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$ openssl verify certificate.crt
C = KE, ST = Nairobi, L = Nairobi, O = MKU, OU = IT, CN = JOB, emailAddress = jobb.roman@gmail.com
error 18 at 0 depth lookup: self-signed certificate
error certificate.crt: verification failed
ayub@ayub-HP-EliteBook-820-G3:~/Desktop$
```

Figure 7: Certificate Testing Results

Reflection

Digital signatures provide authentication, integrity, and non repudiation. The workflow involves signing data with a private key and verifying it using a public key. Certificates validate identities and establish trust. These technologies are widely used in secure email, HTTPS and software distribution.